

Cardiovascular Tests and Endocrine Function Tests



Cardiovascular Function Test

- ❖ Cardiovascular System is responsible for pumping and circulating blood throughout the body.
- ❖ Cardiovascular function test is diagnostic tools to evaluate heart structure and function.
- ❖ • Importance of Tests:
 - Early detection of diseases.
 - Monitoring of chronic conditions.
 - Guiding treatment plans.

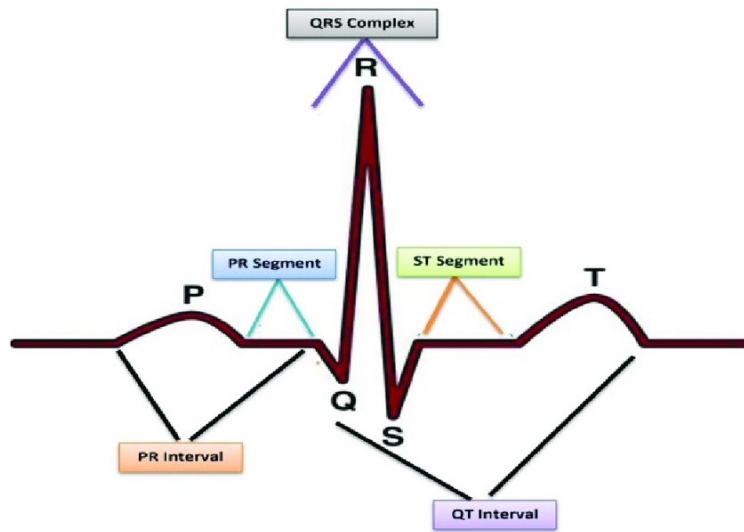
- ❖ key Parameters to be Assessed:
 - Electrical activity (ECG).
 - Blood flow and pressure.
 - Heart muscle and valve function.

❖ **Common cardiovascular tests:**

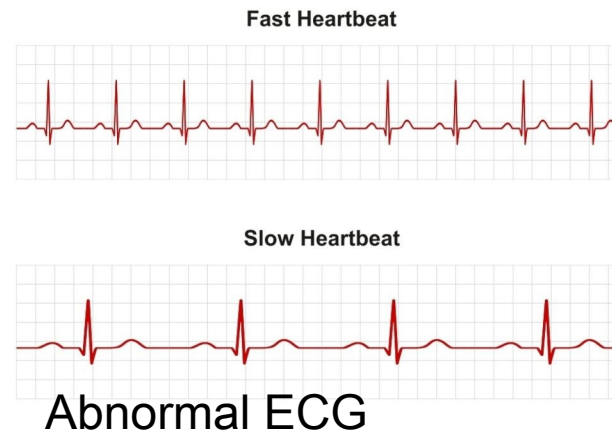
- Electrocardiography (ECG): Records heart's electrical activity.
- Echocardiography: Uses ultrasound to visualize the heart.
- Stress Test: Assesses heart function under physical stress.
- Magnetic resonance imaging (MRI Scans)
- Cardiac Catheterization
- Chest X-rays
- High speed computed axial tomography
- Cardiac enzyme tests

Electrocardiography (ECG)

- ❖ It is most common and most routine test
- ❖ It is very easy and quick, way to assess the heart's function.
- ❖ By this technique the electric activities of the heart studied, detect abnormal cardiac rhythm and myocardium damage
- ❖ Risk associated with ECG are minimal and rare.



Normal ECG



ECG

EKG / ECG RESULTS

NORMAL	ABNORMAL
The heart beats in a normal rhythm for usually 60 and 100 beats /minute.	The heart beat is too slow (less than 60 / minute) The heart beat is too fast (> than 100 beats /minute)
The heart rhythm is regular.	The heart rhythm is not regular.
The tracing looks normal.	The tracing does not look normal.

Diagnosis

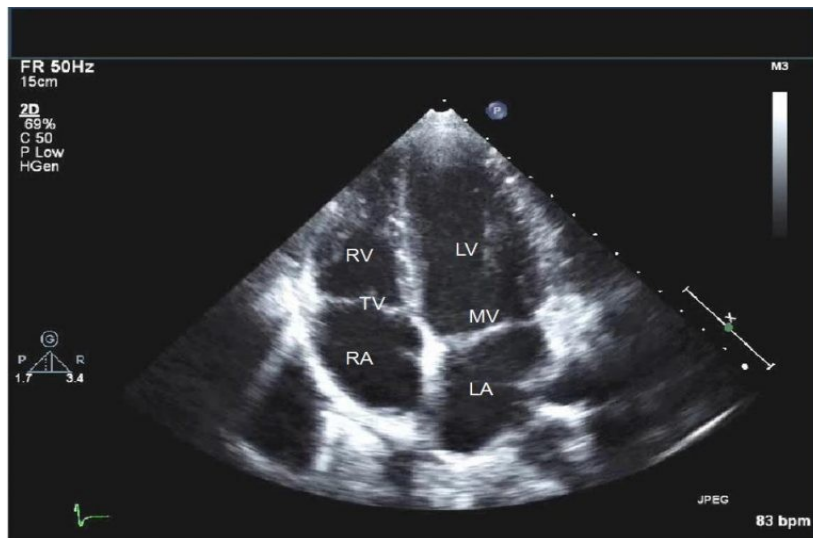
- ❖ To look for cause of chest pain.
- ❖ To Evaluate problems which may be heart-related, dizziness or fainting.
- ❖ To identify the irregular heartbeats.
- ❖ To help determine the overall health of the heart prior to procedures such as heart attack myocardial infraction, endocarditis.
- ❖ To see how an implanted pacemaker is working.
- ❖ To determine how well certain heart medication are working.

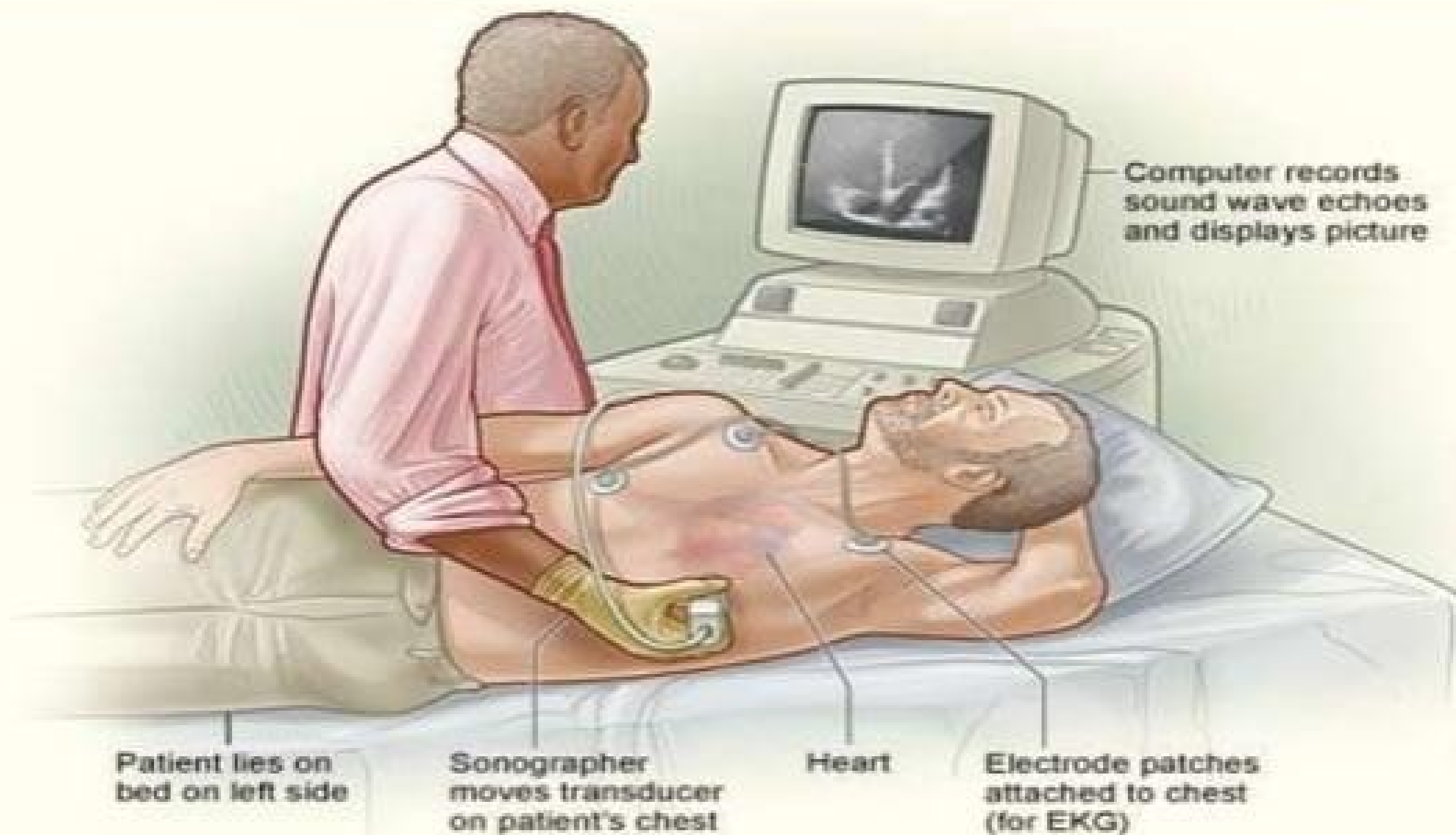
Certain Factors affect or may interfere the result of the ECG

- ❖ Obesity
- ❖ Pregnancy
- ❖ Ascites (fluid build up in the abdomen)
- ❖ Movement during the procedure
- ❖ Certain medications
- ❖ Electrolyte imbalances, such as too much or too little potassium, magnesium, and calcium in the blood.

ECHOCARDIOGRAPHY

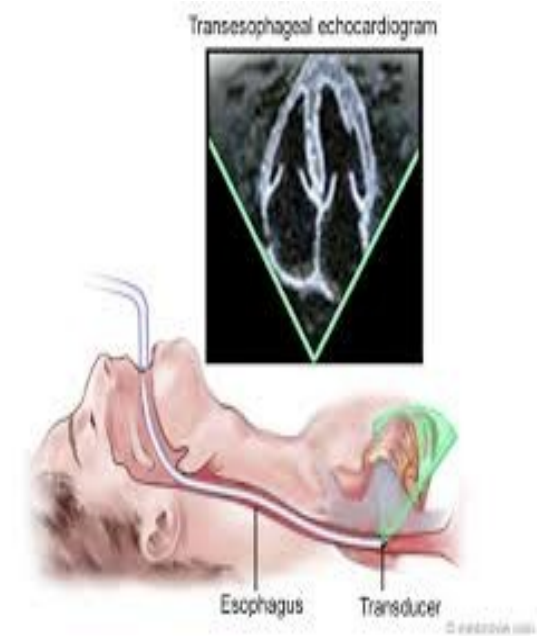
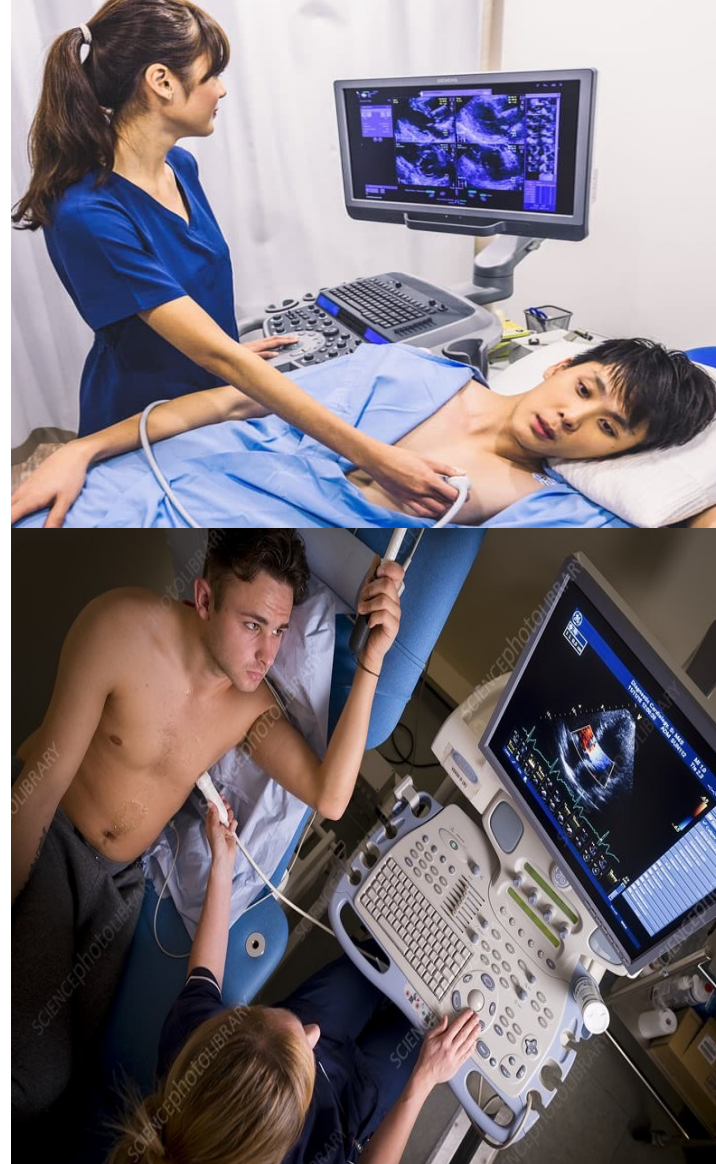
- ❖ An Echocardiography is a test that uses high frequency sound waves (ultrasound) to make pictures /images of heart.
- ❖ This test is also known as echocardiogram/diagnostic cardiac ultrasound.
- ❖ Can evaluate the structures of the heart, as well as the direction of blood flow within it.
- ❖ Produces the images & videos, often using a special probe / transducer that is placed in various places on the chest wall, to view heart from different directions.
- ❖ It is the most common heart tracing done.





TYPES OF ECHOCARDIOGRAPHY

- ❖ Transthoracic Echocardiography
- ❖ Transesophageal Echocardiography
- ❖ Doppler Echocardiography
- ❖ Stress Echocardiography
- ❖ M- mode Echocardiography
- ❖ Color Doppler
- ❖ 2-D Echocardiography
- ❖ 3-D Echocardiography



What is an Echocardiography used to diagnose?

- ❖ Regarding **structure**, the test can assess the general size of the heart,
- ❖ the size of the four heart chambers, and the appearance and function of the four heart valves.
- ❖ It can look at the two septa of the heart; the atrial septum separates the right and left atrium and the ventricular septum separates the right and left ventricles.
- ❖ It can also assess the pericardium (the sac that lines the heart) and the aorta.

- ❖ Regarding **function**, the echocardiogram can determine how the heart valves open and close.
- ❖ It can evaluate whether the heart muscle squeezes appropriately and how efficiently.
- ❖ Cardiac output measures how much blood the heart pumps.
- ❖ Ejection fraction measures what percent of blood within the heart is pumped out to the body with each heartbeat.
- ❖ It can also measure how well the heart relaxes in between beats, when the heart fills for the next pump.
- ❖ Some heart issues that the echocardiogram can help evaluate include the following:
- ❖ Heart valve disorders: Stenosis (narrowed), insufficiency or regurgitation (leaking), and endocarditis (infection of the valves)

- ❖ Abnormalities of the septum: Atrial septal defect, ventricular septal defect, and patent foramen ovale
- ❖ Wall motion abnormalities: Cardiomyopathy, atherosclerotic heart disease (also known as coronary artery disease), and trauma
- ❖ Diseases of the pericardium (the sac that lines the heart): This includes pericardial effusion (assessment of fluid in the pericardial sac)

Magnetic Resonance Imaging (MRI)

This procedure uses a combination of large magnets, radiofrequencies & a computer to make detailed images of organs & structures in your body.

- ❖ Provides detailed picture of heart & blood vessels.
- ❖ Used to evaluate the heart valves & major vessels, detect coronary artery diseases & the extent of damage it has caused, evaluate congenital defects, & detect the presence of tumors or other abnormalities.

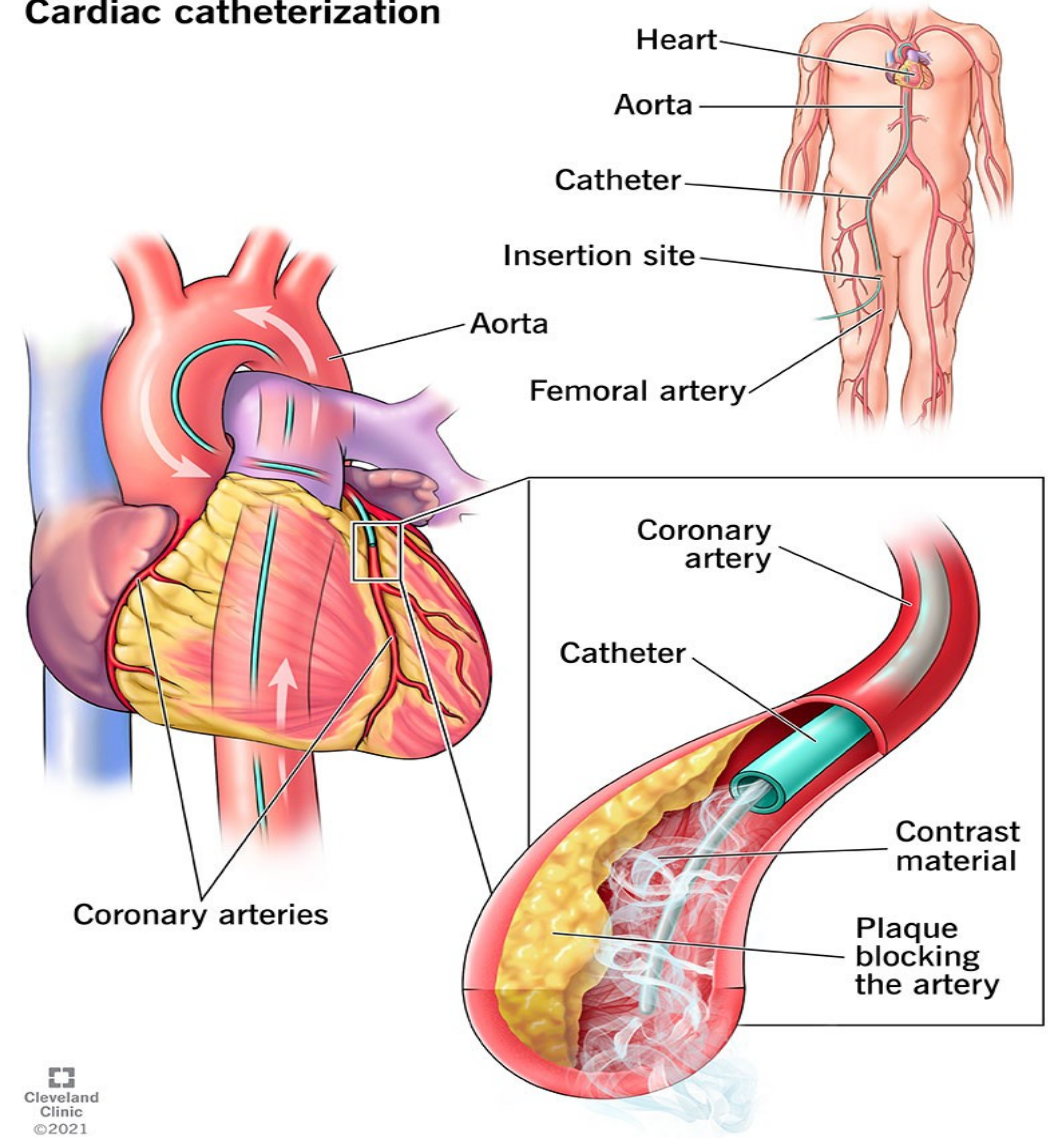


Cardiac Catheterization

It is also called as coronary angiogram.

- It is an invasive procedure.
- Checks the blood flow in the coronary arteries, blood flow, and blood pressure in the chambers of the heart.

Cardiac catheterization



Stress Test

- ❖ Purpose: Evaluates heart performance during exercise or stress.
- ❖ Procedure: Patient walks or runs on a treadmill while heart rate, blood pressure, and ECG are monitored.
 - ❖ Indications:
 - Chest pain evaluation.
 - Detecting coronary artery disease.

- ❖ So many other techniques are available, like
 - High speed computed axial tomography
 - Cardiac enzyme tests
 - Chest X- rays
 - Nuclear imaging

ENDOCRINE FUNCTION TEST

- Endocrine function tests measure the presence and levels of hormones circulating within the body, inside the blood or urine.
- They can vary widely, from straightforward blood serum tests to complex stimulation and suppression tests that assess gland functionality and hormonal balance.
- Understanding these tests is critical as they play a multifunctional role.

Importance of Endocrine function test

- **Diagnosis of Disorders**
- **Monitoring Treatment**
- **Early Detection Explaining Symptoms**
- **Assessing Hormonal Imbalances**
- **Evaluating Gland Function**
- **Preventing Complications**
- **Long-term Health Monitoring**

Thyroid function test

Thyroid Hormones

- Tetraiodothyronine (T4) / Thyroxine (T4)
- Triiodothyronine (T3)

Types of Thyroid Disorders

- Hypothyroidism
- Hyperthyroidism

Specific Tests and Parameters

***Concentration of Thyroid Products**

- Free T4: 0.8 - 2.7 ng/dL -Measures unbound thyroxine in blood
- Total Serum T4: 4 - 12 µg/dL -Measures both bound and free thyroxine.
- Serum T3 Resin Uptake: 25 - 35% -stimates thyroid-binding protein saturation by T3
- Free T4 Index -Reflects active thyroid hormone levels.
- Free T4 index = Total Serum T4 / T3 Resin Uptake.

Refer range: 1.2 - 4.2

- Total Serum T3: 78 - 195 ng/dL Used to detect **T3 toxicosis**.

Hypothalamic-Pituitary-Thyroid Axis Tests

- TSH: 0.3 - 5 μ U/mL
- TRH:Regulates TSH secretion from pituitary

Additional Tests

- Radioactive Iodine Uptake Test: Assess thyroid gland function

Disease	Total serum T4	Total serum T3	T3 resin uptake	Free T4 index	Radioactive iodine uptake test	TSH
Hypothyroidism	↓	↓	↓	↓	↓	↑ 1° ↓ 2°, 3°
Hyperthyroidism	↑	↑	↑	↑	↑	↓
T3 toxicosis	No change	↑	No change	No change	No change	↓
Euthyroid sick syndrome	No change ↓	↓	↑	variable	No change	No change

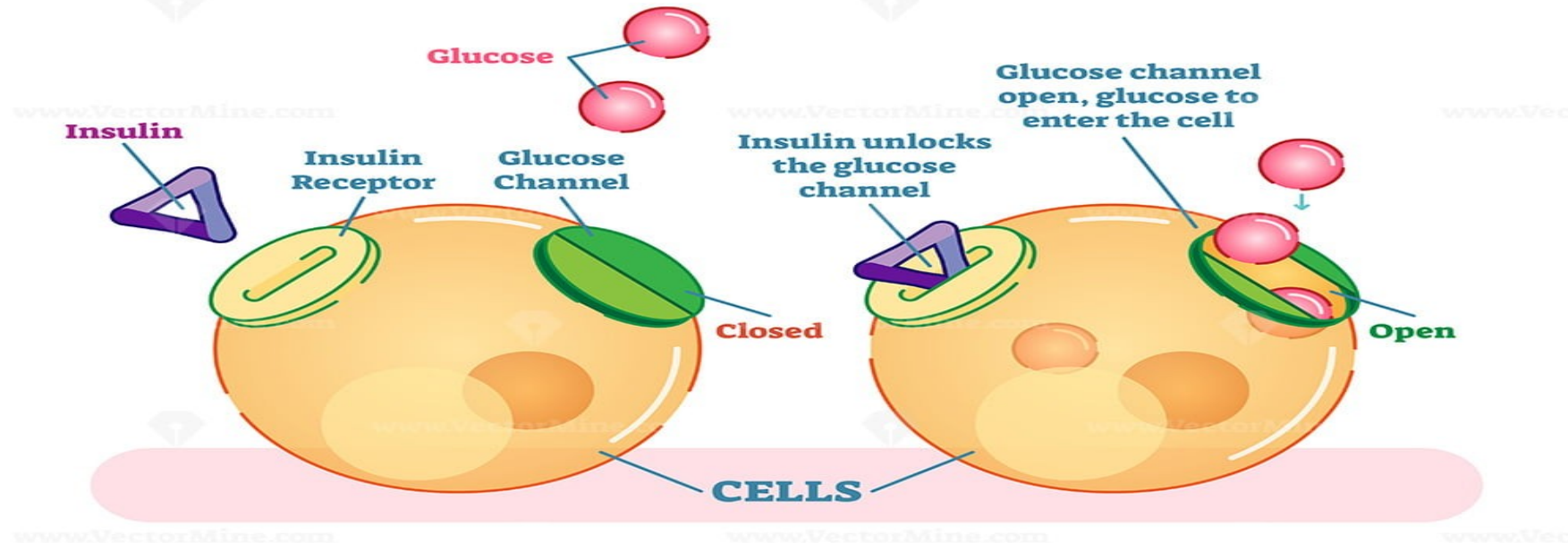
Insulin glucose test

- An insulin glucose test measures the amount of insulin and glucose in your blood.
- Insulin is a hormone that helps move glucose into your cells, where it's used for energy. Glucose is a type of sugar that comes from food.

What is it used for?

- ❖ Find out the cause of hypoglycemia (low blood glucose). This is the main reason for doing insulin testing.
- ❖ Diagnose insulin resistance.
- ❖ Guide treatment decisions for type 2 diabetes.
- ❖ To monitor how well pancreatic islet cell transplantation surgery is working.

HOW DOES INSULIN WORK



Type of insulin test

1.Fasting Insulin Test

Measures insulin levels after fasting for 8-12 hours.It Helps determine insulin sensitivity and is often used in the diagnosis of insulin resistance.

2.Oral Glucose Tolerance Test (OGTT)

Measures blood glucose and insulin levels before and after consuming a glucose-rich drink.

3.Random Insulin Test

Measures insulin levels at any time of the day, regardless of when the patient last ate. Useful for assessing insulin levels in individuals with symptoms of insulin resistance or diabetes.

4.Insulin Tolerance Test

Measures how the body responds to insulin by administering insulin and checking blood glucose levels over time. Evaluates insulin sensitivity and can help diagnose conditions like hypopituitarism or adrenal insufficiency.

Clinical implications of insulin tests

- ❖ Diagnosis of Diabetes: DM-1 and DM-II
by assessing insulin levels and glucose metabolism.
- ❖ High insulin levels in the context of normal glucose levels may indicate insulin resistance, a precursor to type 2 diabetes.
- ❖ Monitoring Treatment Efficacy: Regular insulin testing can help assess how well diabetes treatments (e.g., insulin therapy, oral medications) are working.
- ❖ Results can guide adjustments in medication dosages and dietary recommendations.

Parathyroid hormone function test

- ❖ The main function of the PTH is to regulate Calcium and phosphate level in the blood.
- ❖ A parathyroid function test is a set of blood tests used to assess the function of the parathyroid glands.
- ❖ The main components of a parathyroid function test include:
 - Parathyroid Hormone (PTH) Level
 - Calcium Level
 - Phosphate Level
- ❖ In some cases, additional tests may be performed to further investigate the cause of abnormal PTH levels, including:
 - Vitamin D levels
 - Urinary calcium excretion

Adrenocorticotrophic Hormone Test (ACTH)

- ACTH is Secreted by the pituitary gland and Stimulates the adrenal glands to produce cortisol.

Types of Adrenal Disorders

- Cushing's Syndrome, excess of cortisol
 - Addison's Disease, deficiency of cortisol
 - Ectopic ACTH Syndrome, excess cortisol but released by abnormal source

Specific Tests and Parameters

- Concentration of ACTH: Normal Range: 10 - 60 pg/m

Stimulation Test

- CRH Stimulation Test:- Measures the response of ACTH to CRH.
- ACTH Stimulation Test:- Measures the response of cortisol to synthetic ACTH.

Suppression Tests

- Dexamethasone Suppression Test:- Assesses adrenal gland function by measuring how cortisol levels change in response to dexamethasone.

cortisol test

Types of Cortisol Tests

- Blood Test
- Urine Test
- Saliva Test

Interpretation of Results

- High Cortisol Levels: May indicate Cushing's syndrome, stress, or other health issues.
- Low Cortisol Levels: May suggest Addison's disease or adrenal insufficiency.

Treatment may involve:

- medication or life style change.
- Surgery In cases of tumors affecting adrenal glands.

Growth Hormone (GH) Test

- Produced by the pituitary gland.

1. GH serum test

GH in your body naturally rise and fall throughout the day.

2. Insulin-like growth factor-1 test

3. GH suppression test

- Sweet solution containing glucose is used
- In most people, glucose lowers GH production.

4. GH stimulation test

- A medication that triggers your body to release GH is administered.

Normal Reference Ranges:

- GH: Levels should be less than 5 ng/mL in a random sample; levels rise during childhood and decrease with age.

IGF-1: Varies with age but typically ranges between 100-500 ng/mL in adults.
 below 0.3 nanograms per milliliter (ng/mL) for suppression tests
 peak concentration above 4 ng/mL for stimulation tests in adults
 peak concentration above 5 ng/mL for stimulation tests in children

	Reason for testing	Issues with GH deficiency	Issues with excess GH
Children	GH deficiency or excess GH	• delayed growth and bone development • delayed puberty • below average height	gigantism
Adult	GH deficiency or excess GH	• reduced bone density and muscle mass • abnormal cholesterol	acromegaly

Estrogen, Progesterone, Testosterone

Overview of hormones

- **Estrogen:** Regulates female reproductive system and secondary sexual characteristics, menstrual cycle regulation, fertility, bone health.
- **Progesterone:** Prepares the uterus for pregnancy and regulates menstrual cycle maintains pregnancy, regulates ovulation.
- **Testosterone:** Primary male sex hormone; important for muscle and bone health development of male reproductive tissues, libido.

Testing methods and Normal Ranges

- **Estrogen (Estradiol)** - Serum and urine tests

 - Follicular: 30-100 pg/mL

 - Ovulatory: 100-300 pg/mL

 - Luteal: 200-400 pg/mL

 - Postmenopausal: <30 pg/mL

- **Progesterone** - Serum progesterone levels

 - Follicular: 0.1-1.5 ng/mL

 - Luteal: 5-20 ng/mL

 - Pregnancy: >10 ng/mL

- **Testosterone** - Serum testosterone levels

 - Males: 300-1,000 ng/dL

 - Females: 15-70 ng/dL

Indications for Testing

- **Estrogen** - menstrual irregularities, fertility issues, monitoring HRT, diagnosis of polycystic ovary syndrome(PCOS) or ovarian tumors, investigation of menopausal symptoms.
- **Progesterone** - ovulatory function evaluation, luteal phase defect assessment, early pregnancy viability, reproductive disorder diagnosis(Eg. Ectopic pregnancy).
- **Testosterone** - androgen deficiency symptoms(Eg. Hirsutism, Acne), hyperandrogenism in women, delayed puberty in males, monitoring testosterone therapy.

Common Endocrine Disorders Detected by Endocrine Function Tests

❑ **Hypothyroidism:**

Low thyroid hormone levels (T3/T4) with high TSH. Symptoms: Fatigue, weight gain, cold intolerance.

❑ **Hyperthyroidism:**

High thyroid hormone levels (T3/T4) with low TSH. Symptoms: Weight loss, tachycardia, heat intolerance.

❑ **Diabetes Mellitus:**

Elevated fasting blood glucose and insulin levels, impaired glucose tolerance.

Test: Fasting blood glucose, HbA1c, insulin levels.

❑ **Cushing's Syndrome:**

High cortisol levels. Symptoms: Obesity, round face, high blood pressure, skin thinning.

❑ **Addison's Disease:**

Low cortisol and aldosterone levels. Symptoms: Fatigue, weight loss, low blood pressure.

❑ **Polycystic Ovary Syndrome (PCOS):**

High levels of LH, low levels of FSH, elevated testosterone. Symptoms: Irregular periods, acne, hirsutism.

❑ **Hyperprolactinemia:**

Elevated prolactin. Symptoms: Galactorrhea, infertility, menstrual disturbances.

Factors Affecting Endocrine Function Test Results

A. Biological Factors

- ✓ Age: Hormone levels change with aging (e.g., menopause affects estrogen levels).
- ✓ Gender: Hormonal fluctuations differ in males and females (e.g., menstrual cycle affects estrogen and progesterone levels).
- ✓ Genetics: Some individuals have genetic variations that influence hormone production and metabolism.

B. Physiological Conditions

- ✓ Pregnancy: Increases estrogen, progesterone, and hCG levels.
- ✓ Obesity: Affects insulin, leptin, and cortisol levels.
- ✓ Stress: Elevates cortisol and adrenaline.
- ✓ Time of Day: Some hormones follow a circadian rhythm (e.g., cortisol peaks in the morning).

C. Medications & Lifestyle Factors

Limitations of Endocrine Function Tests

A. Variability in Hormone Levels

- Hormones fluctuate due to circadian rhythm, diet, and stress.
- Single-time-point measurements may not reflect true hormone status.

B. Cross-Reactivity and Assay Interference

- Some tests may have cross-reactivity, leading to false results (e.g., biotin supplements interfere with thyroid function tests).
- Heterophile antibodies can cause false positives in immunoassays.

C. Delayed Diagnosis in Early Disease Stages

- Mild hormone imbalances may not be detected in early disease stages.
- Some disorders require dynamic testing (e.g., ACTH stimulation for adrenal insufficiency).

D. Interpretation Challenges

- Normal ranges vary between labs and populations.
- Results must be correlated with clinical symptoms.

References

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